Experimental Determination of Moment of Inertia

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Contents

1 Abstract

This report investigates the moment of inertia of rotating bodies using experimental measurements of angular velocity as a function of time. Linear regression was used to determine the angular acceleration from the measured data, and the results were compared for different experimental configurations. The aim was to examine how the distribution of mass affects the moment of inertia and to evaluate how well the experimental results agree with theoretical expectations.

2 Introduction

Moment of inertia is a physical property of anything which rotates, charachterized by the resistance to changes in angular velocity. Since this affects how much force needs to be applied to accelerate or decelerate objects, knowing how to calculate anc measure it is crucial for many applications. In this study several means of measuring the moment of inertia practically were explored and compared with theoretical values. The aim was to determine if this was a valid means of mesuring the moment of inertia and how accurate of a method it was. A second goal of the experiment was to determine the effect of impulse on a rotating system to show with a practical example that angular momentum is conserved in systems like this while angular kinetic energy is not.

3 Theory/Background

To calculate with rotational movement and can not idealize the mass to a particle as often done in translational calculations, you have to use expresions giving a link between the mass of an object and how it is distributed [**up**]

$I = moment of inertia$, different geometric forms will give diferent calculations about its centroid being the center of the distributed mass.

Examples of equations of moment of inertia

Equation for a circular plate

$$I_x = I_y = m\frac{r^2}{2} \tag{1}$$

[**up**]

Equation for a circular ring

$$I_x = I_y = \frac{1}{2}m(r_1^2 + r_2^2) \tag{2}$$

[**up**]

Using this link Newtons second law $\sum F = ma$ analogusly can be aplied for rotational movement [**up**].

$$\sum \tau = I\alpha \tag{3}$$

τ equals torsion I equals moment of inertia and α the angular acceleration

To calculate Torque (τ) it is the distance to where the aplied Force perpendicular to the rotational center.

$$\tau = Tr \tag{4}$$

[**ph**]

T is the aplied force in Newton and r is the distance to the point of perpendicular aplied force.

Given an expresion for $\tau = Tr$ the two formulas can be joined to

$$Tr = I\alpha \tag{5}$$

In the case that the force is generated with a free faling weight the folowing substitutions can be made $\sum F = mg - T = ma$. a = the linear acceleration of the falling mass giving the expression.

$$mgr = I\alpha \tag{6}$$

When the centroid of an object and the axes of rotation do not align The Paralell axel Theorem (Steiners's Theorem) can be used to calculate the moment of inertia.

$$I_{total} = I_{cm} + mr^2 \; (r = the\, distance\, between\, axis) \tag{7}$$

[**ph**]

With the given equations a expression for the experimantal moment of inertia can be derived.

$$I = \frac{mgr}{\alpha} - mr^2 \tag{8}$$

For calculations of angular mommentum L and rotational energy k_{rot} fomulas analogus to straight movement are used.

$$E_{rot} = \frac{1}{2}\omega I \tag{9}$$

$$L = I\omega \tag{10}$$

[**ph**]

For error estimation the standard deviation can be calculated to give a sense of how close to the mean value the mesurements are distributed. x_i is the experimental value, x the mean value and n the amount of data collections

$$S = \sqrt{\frac{\sum_{i=1}^{n}(x_i - x)^2}{n-1}} \tag{11}$$

[**ph**]

4 Method

Two differrent experiments were conducted one to experimentaly decide the moment of inertia and one to see what hapens with angular momenton and rotational energy after an impact with an aditional body folowing allong with the rotation.

Both experiments were conducted using the PASCO Capstone system with the sensor setup shown in figure **??**

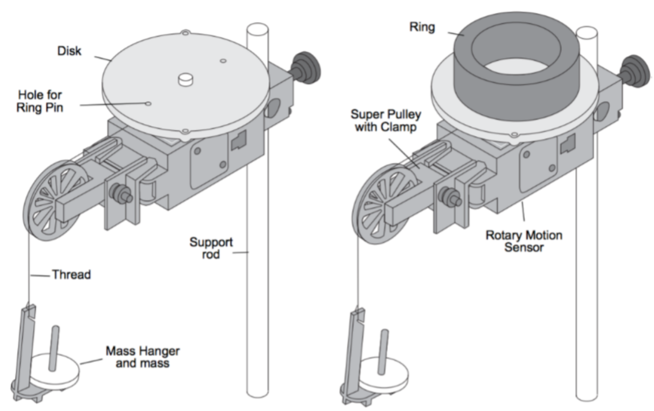

Figure 1: Graphic showing the sensor setup system.

The data was logged with angular velocity and time on the y and x axis in the Pasco system. to have controlled force aplied in the system a smal weight atached to the disk in the sensor system was used. The weight was realesed and the PASCO system recoded the event, for the first experiment the flyweel alone was measured five times and with an aditional ring weight added to the system as in figure **??**

5 Results

Below is a table showing direct measurements of the relevant properies of the experiment setup for calculating the moment of inertia. Masses were measured with scales and distances and lengths with calipers.

Entity	Value	Unit
m_{disk}	119,63	g
m_{ring}	457,26	g
m_{weight}	9,66	g
r_{disk}	47,5	mm
$r_{ring\ outside}$	27	mm
$r_{ring\ inside}$	38,5	mm
$r_{atachment\ point}$	23,75	mm

Table 1: Mesurements for ring, disk and atachment point

	Disk(rad/s^2)	Disk+ring(rad/s^2)
1	14,0	2,99
2	13,9	3,02
3	14,1	3,02
4	14,1	3,05
5	14,1	3,04
M_α	14,04	3,022
S_α	0,089	0,0238

Table 2: Data points for experiment 1

	Disk(rad/s)	Disk+ring(rad/s)	Displacement(mm)
1	17,554	4,26	6
2	17,606	4,32	8
3	15,093	3,757	4
4	21,271	4,725	9
5	16,153	4,725	5
M_ω	17,53	4,095	6,4
S_ω	2,336	0,498	

Table 3: Data points for experiment 2

Listed below is the plotted data exported from the CASPECO software showing the experiemnts. Note that the scale on the axis are not the same between the figures.

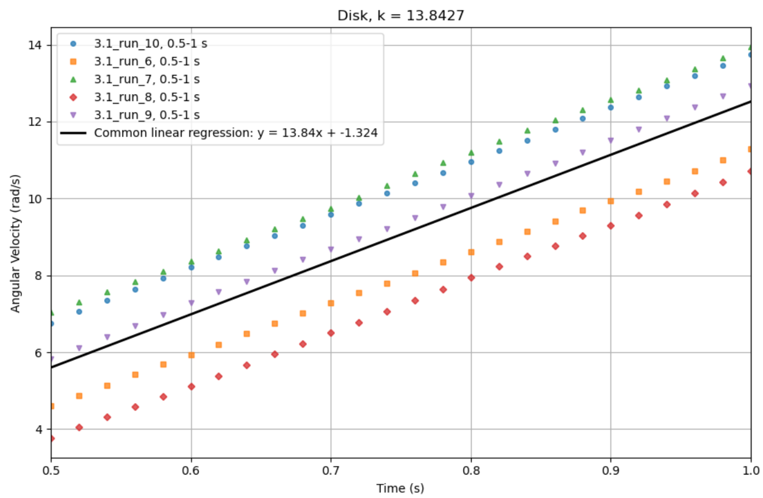

Figure 2: Graph from the experiment used to determine the moment of inertia of a disk. The k-value gives the angular acceleration.

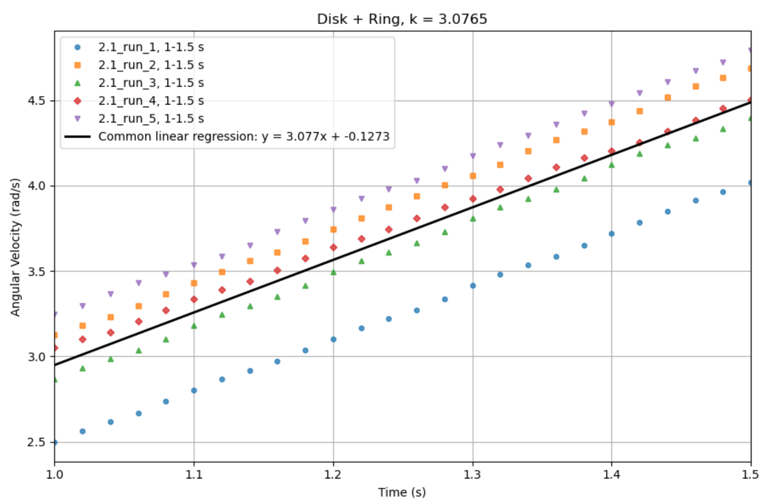

Figure 3: Graph from the experiment used to determine the moment of inertia of a disk with added weight. The k-value gives the angular acceleration.

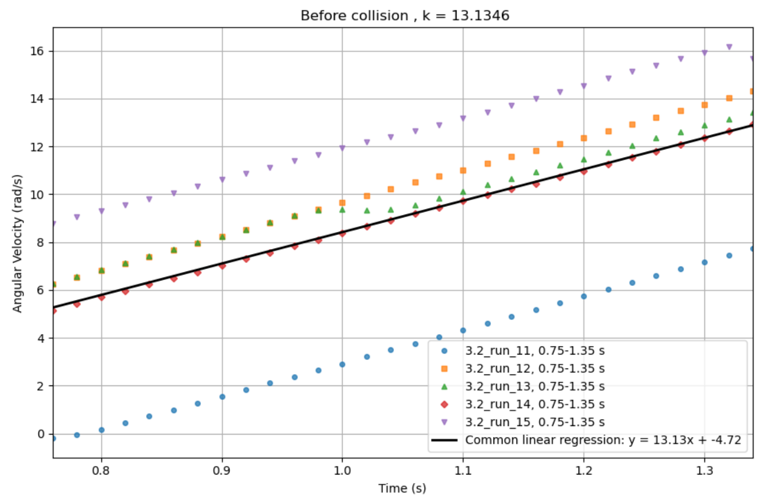

Figure 4: Graph showing the angular speed before impact of the added weight. The k-value gives the angular acceleration.

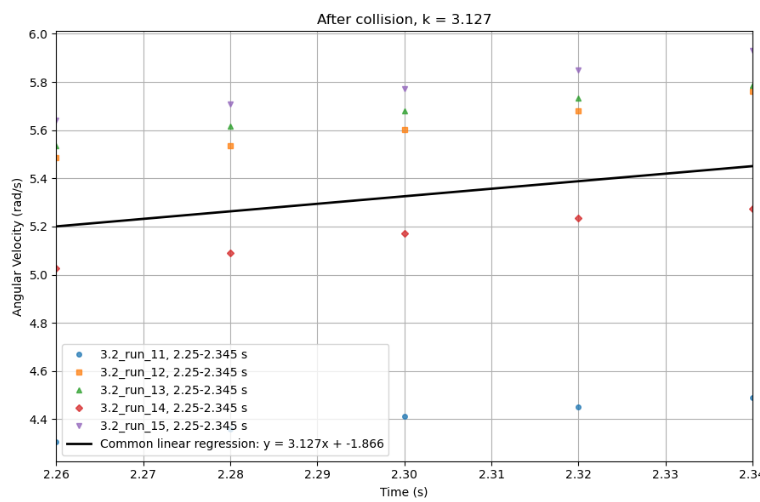

Figure 5: Graph showing the angular speed after impact of the added weight. The k-value gives the angular acceleration.

Experiment	Mean(α)	Standard deviation
$Experimental\ I_{disk}$	14,04	0,089
$Experimental\ I_{disk+ring}$	3,022	0,0238

Table 4: Standard deviation calculated acording to equation: (**??**)

Experiment	Mean(ω)	Standard deviation
$Experimental\ L/K_{pre-impact}$	17,53	2,336
$Experimental\ L/K_{post-impact}$	4,095	0,498

Table 5: Standard deviation calculated acording to equation: (**??**)

haj

6 Calculations

Using equation:(**??**) the experimental moment of inertia can be calculated only knowing the mass of the faling weight and the radius of the pulling disk.

$$I_{disk} = \frac{0,00966kg \cdot 9,81\frac{rad}{s^2} \cdot 0,02375m}{14,04\frac{rad}{s^2}} - 0,00966 \cdot 0,02375m^2 = 1,548kgm^2 \cdot 10^{-4} \tag{12}$$

$$I_{disk+ring} = \frac{0,00966kg \cdot 9,81\frac{rad}{s^2} \cdot 0,02375m}{3,022\frac{rad}{s^2}} - 0,00966 \cdot 0,02375m^2 = 7,393kgm^2 \cdot 10^{-4} \tag{13}$$

To calculate the theroretical values for the disk equation: (**??**) are used and for the disk and the ring it is combined with equation:(**??**)

$$I_{disk} = \frac{0,11963kg \cdot 0,0475^2m}{2} = 1,349kgm^2 \cdot 10^{-4} \tag{14}$$

$$\begin{aligned} I_{disk+ring} = {} & \frac{0,11963kg \cdot 0,0475^2m}{2} \\ & + \frac{0,455726kg \cdot (0,027^2m + 0,0385^2m)}{2} = 6,388kgm^2 \cdot 10^{-4} \end{aligned} \tag{15}$$

Before calculating the kinetic energy and the angular momentum, the agular moment has t be calculated, to calculate with the paralell theorem the mean displacement are used (6,4mm).

$$I_{before} = \frac{0,11963kg \cdot 0,0475^2m}{2} = 1,548kgm^2 \cdot 10^{-4} \tag{16}$$

$$\begin{aligned} I_{after} = {} & \frac{0,11963kg \cdot 0,0475^2m}{2} + \frac{0,455726kg \cdot (0,027^2m + 0,0385^2m)}{2} + \\ & \frac{0,19963kg \cdot 0,0064^2m}{2} = 6,429kgm^2 \cdot 10^{-4} \end{aligned} \tag{17}$$

To calculate and compare the Kineticenergy before and after the impact equation: (**??**) are used

$$E_{disk} = \frac{17,53\frac{rad}{s} \cdot 1,548 \cdot 10^{-4}kgm^2}{2} = 0,001356kgm^2/s^2 \tag{18}$$

$$E_{disk+ring} = \frac{4,095\frac{rad}{s} \cdot 6,429 \cdot 10^{-4}kgm^2}{2} = 0,001316kgm^2/s^2 \tag{19}$$

To calculate and compare the angular momentum before and after the impact equation: (**??**) are used

$$L_{disk} = 1,548 \cdot 10^{-4} kgm^2 \cdot 17,53 rad/s = 0,002713 Kgm^2/s \tag{20}$$

$$L_{disk+ring} = 6,429 \cdot 10^{-4} kgm^2 \cdot 4,095 rad/s = 0,002632 kgm^2/s \tag{21}$$

To compare the before and after impact the procentual change are calculated

$$L_{\%} = \frac{L_e - L_f}{L_f} \cdot 100 = 2,985\% \tag{22}$$

$$K_{\%} = \frac{K_e - K_f}{K_f} \cdot 100 = 2,949\% \tag{23}$$

7 Discussion

All the data show good linearity, indicating that the use of a linear model was indeed appropriate, while forces such as air resistance are exponential in nature they were not of a sufficient magnitude to significantly affect the results. Additionally, none of the data strayed from the theoretical values by completely unexpected amounts such as an order of magnitude. The observed moments of inertia did however also not completely align with the expected values, differing by at 1.005 $kgm^2 \cdot 10^{-4}$ which is not insignificant when the observed value is only $7.393 kgm^2 \cdot 10^{-4}$.

Wether this deviance is to be considered significant depends on the application. If for instance the aim is to ensure a motor in a stationary application has sufficient torque, this method is totally adequate as the resulting moment of inertia is higher than the theoretical one. Dimensioning with these experimental values should therefore always ensure some limited level of safety marigin, without further testing, this should however not be relied upon and instead be used in conjunction with additinal safety marigins.

8 References

9 Appendix